Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper Aug-2024

PART - A

(25 Marks)

1.a)	Write the consequences of distributed systems.	[2]
b)	What is Internet-scale distributed systems?	[3]
c)	Brief about clock skew and clock drift.	[2]
d)	Define total and causal ordering of multicast messages.	[3]
e)	Differentiate synchronous and asynchronous communication.	[2]
f)	Write about At-least-once semantics and At-most-once semantics.	[3]
g)	List any four UNIX file system operations.	[2]
h)	What is URI, URN and URL?	[3]
i)	Write about ACID properties of a transaction.	[2]
j)	Brief about strict two-phase locking.	[3]

PART - B

(50 Marks)

2.a) What is client-server computing? Explain remote invocation in this context of client-server computing.
b) What are the principle characteristics of Distributed Systems? [5+5]

OR

3.a) What are the performance bottlenecks of distributed systems with respect to scalability? Discuss the solution to avoid these bottlenecks.
b) What are the two variants of the interaction model in distributed systems? Discuss the differences between these variants differ. [5+5]

4.a) Explain Cristian's method for synchronizing clocks.
b) Write and discuss Maekawa's voting algorithm. [5+5]

OR

5.a) Write the 'snapshot' algorithm proposed by Chandy and Lamport.
b) Explain the Ricart and Agrawala's multicast-based mutual exclusion algorithm in brief. [5+5]

6.a) Explain about TCP stream communication.
b) Discuss an overlay network used in Skype. [5+5]

OR

7.a) What type of failures may occur in IP multicast? How it can be resolved?
b) Explain request-reply protocol with respect to HTTP. [5+5]

8.a) Elaborate various distributed file system requirements.
b) What are the design and implementation issues of distributed shared memory? [5+5]

OR

9.a) Discuss about sequential consistency in DSM system.
b) How the DNS is implemented using Berkeley Internet Name Domain (BIND)? [5+5]

10.a) What is the need for concurrency control? Explain with an example.
b) How deadlocks are detected in distributed environment? Explain. [5+5]

OR

11.a) Write about optimistic concurrency control.
b) Explain about log based recovery in distributed environment. [5+5]

---ooOoo---

Used paper Aug-2024